

① How many solutions has the problem?

a) $x'' + t^2 x = 0, x(0) = 0$

$x(t) = c_1 \cdot \dots$

b) $x'' + t^2 x = 0, x(0) = 0, x'(0) = 0.$

c) $x'' + t^2 x = 0, x(0) = 0, x'(0) = 0, x''(0) = 1.$

$x'' + t^2 x = 0 \rightarrow$ this is a second order linear d.e. with a non-constant coeff.

Pb. b) is an IVP. Applying the existence and uniqueness theorem for the IVP, we deduce that b) has a unique sol.

Note that $x=0$ is a solution of b) $\xrightarrow[\text{unig. and exist.}]{\text{th. of}}$ $x=0$ is the unique sol.

c) $x=0$ is the unique function that verifies the first 3 conditions in pb c). Since this function does not verify the last condition, we deduce that pb c) has no sol.

Another method: assume, by contradiction, that pb c) has a solution

$\ell: \mathbb{R} \rightarrow \mathbb{R}$. Then $\ell''(t) + t^2 \ell(t) = 0, (\forall) t \in \mathbb{R}$

$\ell(0) = 0$

$\ell'(0) = 0$

$\ell''(0) = 1.$

We check for $t=0 \Rightarrow \ell''(0) + 0 \cdot \ell'(0) = 0.$

$\Rightarrow 1 + 0 = 0$ ctr.

a) $x'' + t^2 x = 0, x(0) = 0$

Let $\eta \in \mathbb{R}$ and we consider the IVP $x'' + t^2 x = 0, x(0) = 0$ and $x'(0) = \eta.$

This IVP has a unique sol. $\ell_\eta: \mathbb{R} \rightarrow \mathbb{R}$

Then pb. a) has an entire family of solutions $\ell_\eta: \mathbb{R} \rightarrow \mathbb{R}, (\forall) \eta \in \mathbb{R}$

②

② $v \in C^1(\mathbb{R})$, $v(0)=0$, $v'(0) \neq 0$, $xv(x) > 0$, $(\forall) x \in \mathbb{R}^*$

$$\begin{cases} \dot{x} = y \\ \dot{y} = -v(x) \end{cases}$$

a) first integral

b) stability of the equil. point $(0,0)$.

$$\frac{dy}{dx} = \frac{-v(x)}{y} \Rightarrow y dy = -v(x) dx \quad \Big| \int$$

$$\Rightarrow \int_0^y y dy = \int_0^x -v(s) ds + C$$

$$\Rightarrow \frac{1}{2} y^2 = - \int_0^x v(s) ds + C$$

$$H(x,y) = \frac{1}{2} y^2 + \int_0^x v(s) ds$$

$$H: \mathbb{R}^2 \rightarrow \mathbb{R}^*$$

$$\nabla H(x,y) = (v(x); y)$$

$$y \cdot \frac{\partial H}{\partial x}(x,y) - v(x) \cdot \frac{\partial H}{\partial y}(x,y) = 0, \quad (\forall) (x,y) \in \mathbb{R}^2 \quad (\Rightarrow)$$

$$(\Rightarrow) y \cdot v(x) - v(x) \cdot y = 0, \quad (\forall) (x,y)$$

$$\nabla H(x,y) = 0 \quad (\Rightarrow) \begin{cases} v(x) = 0 \\ y = 0 \end{cases}$$

$\Rightarrow H$ has locally

$$\Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

$(0,0)$ the only critical point of H .

b) $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x,y) = \begin{bmatrix} y \\ -v(x) \end{bmatrix}$, $f(0) = (0,0) \Rightarrow (0,0)$ equil.

$$Jf(x,y) = \begin{bmatrix} 0 & 1 \\ -v'(x) & 0 \end{bmatrix} =$$

$$Jf(0,0) = \begin{bmatrix} 0 & 1 \\ -v'(0) & 0 \end{bmatrix}$$

$$\det(Jf(0,0) - \lambda I_2) = \lambda^2 + v'(0) = 0.$$

$$xv(x) > 0, (\forall) x \in \mathbb{R}^* \in (-\infty, 0) \Rightarrow v(x) < 0, (\forall) x \in (-\infty, 0) \quad (\Rightarrow)$$

$$xv(x) > 0, (\forall) x \in (0, \infty) \Rightarrow v(x) > 0, (\forall) x \in (0, \infty)$$

$$\Rightarrow \exists v \in$$

$$\lambda_{\mathbb{R}} = \pm i \sqrt{v'(0)}$$

$$2) \begin{cases} \dot{x} = x(1-y) \\ \dot{y} = -y(2-x) \end{cases}, x \in \mathbb{R}$$

a) first integral; check first order partial d.e.

b) $(1, 1)$ hyperbolic point.

$$a) \frac{\dot{y}}{\dot{x}} = \frac{dy}{dx} \Rightarrow \frac{-y(2-x)}{x(1-y)} = \frac{dy}{dx} \Rightarrow \frac{1-y}{-y} dy = \frac{2-x}{x} dx \Rightarrow$$

$$\Rightarrow \left(1 - \frac{1}{y}\right) dy = \left(\frac{2}{x} - 1\right) dx \quad \left| \int \right.$$

$$\Rightarrow \int \left(1 - \frac{1}{y}\right) dy = \int \left(\frac{2}{x} - 1\right) dx$$

$$\Rightarrow y - \ln y = 2 \ln x - x + C$$

$$\Rightarrow y + x - \ln y - 2 \ln x = C.$$

$$\underbrace{\hspace{10em}}_{H(x,y)}$$

$$H: (0, \infty) \times (0, \infty) \rightarrow \mathbb{R}$$

$$\nabla H(x,y) = \left(\frac{\partial H}{\partial x}, \frac{\partial H}{\partial y} \right) = \left(1 - \frac{2}{x}, 1 - \frac{1}{y} \right)$$

$$\frac{\partial H}{\partial x} \dot{x} + \frac{\partial H}{\partial y} \dot{y} = 0, \quad (x,y) \in (0, \infty) \times (0, \infty)$$

$$\Rightarrow \left(1 - \frac{2}{x}\right) x(1-y) + \left(1 - \frac{1}{y}\right) (-y)(2-x) = 0$$

$$\Rightarrow (1-y)(x-2+2-x) = 0 \quad \checkmark$$

$$\nabla H(x,y) = (0,0) \Leftrightarrow \begin{cases} 1 - \frac{2}{x} = 0 \\ 1 - \frac{1}{y} = 0 \end{cases} \Leftrightarrow \begin{cases} x-2=0 \\ y-1=0 \end{cases} \Rightarrow \begin{cases} x=2 \\ y=1 \end{cases} \Rightarrow (2,1) \text{ unig. critical point.}$$

$\Rightarrow H$ not locally ct.

H first integral in $(0, \infty) \times (0, \infty)$.

$$f(x,y) = \begin{bmatrix} x - xy \\ -2y + xy \end{bmatrix}, \quad f(2,1) = \begin{bmatrix} 2-2 \\ -2+2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow (2,1) \text{ equil. point.}$$

$$J(x,y) = \begin{bmatrix} 1-y & -x \\ y & -2+x \end{bmatrix} \Rightarrow J(2,1) = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$$

$(2,1)$ not hyp. pt.
as $P_2(\lambda) = P_1(\lambda_2) = 0$

$$\det(J(2,1) - \lambda I) = 0 \Rightarrow \begin{vmatrix} -\lambda & -2 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + 2 = 0 \Rightarrow \lambda = \pm i\sqrt{2}$$

(3)

2!

(12) Barbălat's Lemma.

 $\varphi: \mathbb{R} \rightarrow \mathbb{R}$, a C^1 function.

$$\exists \lim_{t \rightarrow \infty} \varphi(t), \lim_{t \rightarrow \infty} \varphi'(t) \text{ finite} \xRightarrow{?} \lim_{t \rightarrow \infty} \varphi'(t) = 0.$$

Mean Value Theorem: $\exists \varrho_n \in [n, n+1]$, s.t. $\frac{\varphi(n+1) - \varphi(n)}{n+1 - n} = \varphi'(\varrho_n)$

$$\Rightarrow \varphi(n+1) - \varphi(n) = \varphi'(\varrho_n)$$

$$\left. \begin{array}{l} \exists \lim_{t \rightarrow \infty} \varphi(t), \text{ finite.} \Rightarrow \lim_{n \rightarrow \infty} \varphi(n) = L \\ \text{and} \\ \lim_{n \rightarrow \infty} \varphi(n+1) = L. \end{array} \right\} \Rightarrow (-) \Rightarrow \lim_{n \rightarrow \infty} \varphi(n+1) - \lim_{n \rightarrow \infty} \varphi(n) = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \varphi'(\varrho_n) = 0.$$

$$\left. \begin{array}{l} \lim_{n \rightarrow \infty} \varrho_n = \infty \\ \exists \lim_{t \rightarrow \infty} \varphi'(t) = L' \end{array} \right\} \Rightarrow \lim_{n \rightarrow \infty} \varphi'(\varrho_n) = L' = 0 \Rightarrow \lim_{t \rightarrow \infty} \varphi'(t) = 0.$$

(13) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a C^1 function and $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be a solution of the diff. eq. $\dot{x} = f(x)$ and s.t. $\lim_{t \rightarrow \infty} \varphi(t) = 2$. Prove that 2 is an equilibrium point of $\dot{x} = f(x)$.

$$\varphi \text{ is a solution} \Rightarrow \varphi'(t) = f(\varphi(t)).$$

$$\text{We know } \lim_{t \rightarrow \infty} \varphi(t) = 2 \in \mathbb{R} \quad \left. \begin{array}{l} \varphi \text{ - cont.} \end{array} \right\} \Rightarrow \lim_{t \rightarrow \infty} f(\varphi(t)) = f(2)$$

$$\varphi \text{ is a solution of the equation } \dot{x} = f(x) \Rightarrow \varphi'(t) = f(\varphi(t)), (\forall t \in \mathbb{R})$$

$$\text{We know } \lim_{t \rightarrow \infty} \varphi(t) = 2 \in \mathbb{R} \quad \left. \begin{array}{l} \varphi \text{ is a } C^1 \text{ function cont.} \end{array} \right\} \Rightarrow \lim_{t \rightarrow \infty} f(\varphi(t)) = f(2)$$

$$\Rightarrow \lim_{t \rightarrow \infty} \varphi'(t) = f(2).$$

$$\text{We apply Barbălat's Lemma} \Rightarrow \lim_{t \rightarrow \infty} \varphi'(t) = 0 \quad \left. \begin{array}{l} \end{array} \right\} \Rightarrow f(2) = 0 \Rightarrow 2 \text{ equil. point}$$